Wu 2020 AML — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

One connected kNN component at every θ (2/10/20); 488 cells (0.5%) in tiny detached specks, removed. Harmony θ=10 minimises residual same-sample enrichment (2.7×; θ=20 re-segregates). iLISI 9.5/36 (26% of ceiling). Decisive: between-sample R^2 of the embedding = 0.01 — patient identity explains ~1% of manifold structure, so this is an integrated continuum, NOT a per-patient clonal archipelago (cf. Liu2021 MM).

Gate 2 — module geography → NO-GO

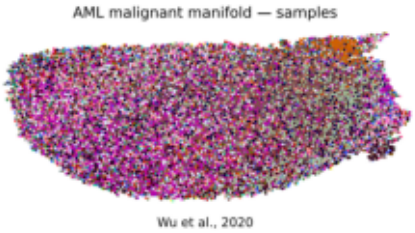
Embryonic module pattern ABSENT. Smoothed signature geography is featureless (score std ~0.007; no Gast/pSom poles, no Crest gradient); module means ~0 (Gast +0.002, pSom +0.007, Crest −0.004). Gast and pSom ANTI-correlate (smoothed r=−0.37) instead of overlapping as the non-branching case requires. No developmental axis: modules unaligned with any diffusion component (|r|≤0.14), flat along pseudotime, and the Crest "terminal" position FLIPS with the root choice (0.59 vs 0.41) — a rooting artifact, not biology.

Gate 3 — cross-repression → NA

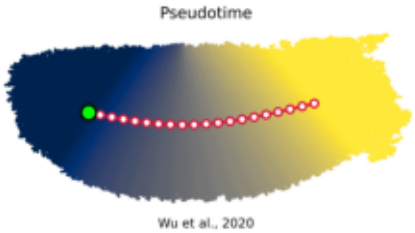
Not applicable — no bifurcation (Gate 2 NO-GO). Branching-only 04_correlation / 05_modules not produced.

STEP FIGURES

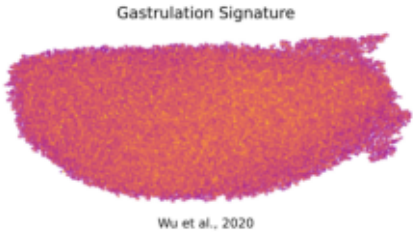
Manifold — samples



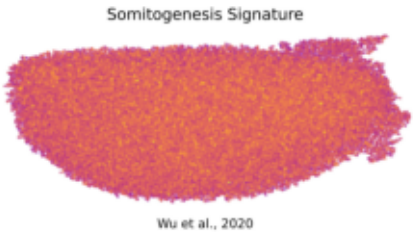
Pseudotime



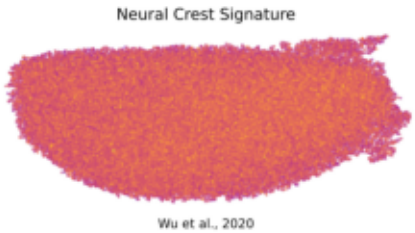
Gastrulation



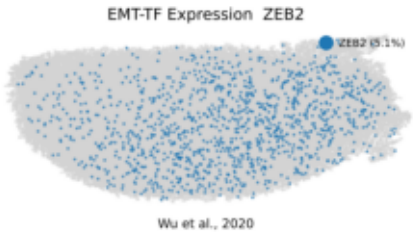
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Bone-marrow acute myeloid leukemia (Wu et al., 2020), Microwell-seq — 96,413 malignant blasts across 36 samples. A trajectory is fit, but the embryonic module pattern is absent (Gate 2 NO-GO; pattern-absent).

Verdict.

The embryonic developmental pattern does NOT reproduce — malignant AML blasts form an integrated, continuous manifold with no Gast/pSom hub, no arms, and no terminal Crest gradient (Gate 2 NO-GO). Geometry: none (pattern absent). Corrects three prior attempts that read a union-control-pool scoring artifact off an arbitrary root on a featureless blob. Gate 1 GO (integrated continuum, between-sample R^2=0.01), Gate 3 NA.

Parameter choices

Min-cell threshold	≥100 (clean gap 98→417, 4.26×) → 36/38 samples, 96,901 cells	**Integration**	Harmony θ=10 (residual-enrichment min); n_pcs=20, k=30, spread=1.3; 443 HVG
Island removal	radius-graph 6× median, keep largest → 488 removed, 96,413 final	**Trajectory**	scFates ElPiGraph curve (N=20, λ=0.3, μ=100); non-branching; root Gast/pSom-high tip
Gate 2 verdict	pattern absent — module means ~0, geography featureless, Gast–pSom anti-correlate (−0.37)	**Rooting artifact**	Crest terminal position flips with root (0.59 tip0 vs 0.41 tip2)
EMT-TFs	panel SNAI/ZEB/TWIST/FOXC/PRRX (VIM excluded); only ZEB2 (5.1%) clears 3% floor		